05. Windows Basics for Cybersecurity

1. Processes

Definition: A process is a running instance of a program.

Examples:

- explorer.exe — Windows graphical shell.
- notepad.exe — Notepad application.
- powershell.exe — PowerShell command-line environment.

Key points:

- Processes use CPU, memory and other resources.
- A process has a process ID (PID).
- Analysts investigate unexpected processes, parent-child relationships and suspicious command-line arguments.

2. Services

Definition: A Windows service is a program designed to run in the background and provide system or application functionality.

Examples:

- Windows Update service.
- Windows Defender-related services.
- Print Spooler service.

Analysts should investigate unexpected services, unusual executable paths and services created without authorization.

Useful PowerShell command:
Get-Service

3. Windows Registry

Definition: The Windows Registry is a hierarchical database containing configuration settings for Windows, applications, users and devices.

Important root keys:

- HKEY_LOCAL_MACHINE (HKLM) — system-wide settings.
- HKEY_CURRENT_USER (HKCU) — settings for the current user.
- HKEY_USERS (HKU) — settings for loaded user profiles.
- HKEY_CLASSES_ROOT (HKCR) — file associations and related class information.
- HKEY_CURRENT_CONFIG (HKCC) — current hardware profile information.

Analysts may examine registry changes associated with persistence, software installation or configuration changes.

4. Users and Groups

Definition: A user account represents an identity that can sign in or access resources. A group collects accounts or other security principals so permissions can be assigned together.

Examples:

- Standard user — limited permissions.
- Administrator — elevated system-management permissions.
- Administrators group — accounts with extensive local administrative rights.

Security principle: use the least privilege necessary for each task.

5. User Account Control (UAC)

Definition: UAC is a Windows security feature that helps prevent unauthorized changes by requesting consent or administrator credentials for elevated operations.

Key points:

- Helps reduce accidental or unauthorized administrative changes.
- An administrator account may still run programs without full elevation until elevation is approved.
- Unexpected elevation prompts should be investigated.

6. Three Main Event Viewer Logs

Log	Purpose	Example
Application	Records events from applications and software components	Application errors
Security	Records security-related activity according to audit policy	Successful and failed logons
System	Records Windows system components and service events	Driver or service events

Open Event Viewer by running:
eventvwr.msc

Important note: the events recorded depend on Windows configuration, auditing policy, permissions and log retention.

7. Useful Windows Commands

List processes:
Get-Process

List services:
Get-Service

List local users:
Get-LocalUser

List local groups:
Get-LocalGroup

Show current identity:
whoami

8. Conclusion

Understanding processes, services, registry settings, users, groups and Windows event logs helps a SOC analyst investigate suspicious activity, unauthorized changes and potential compromise.